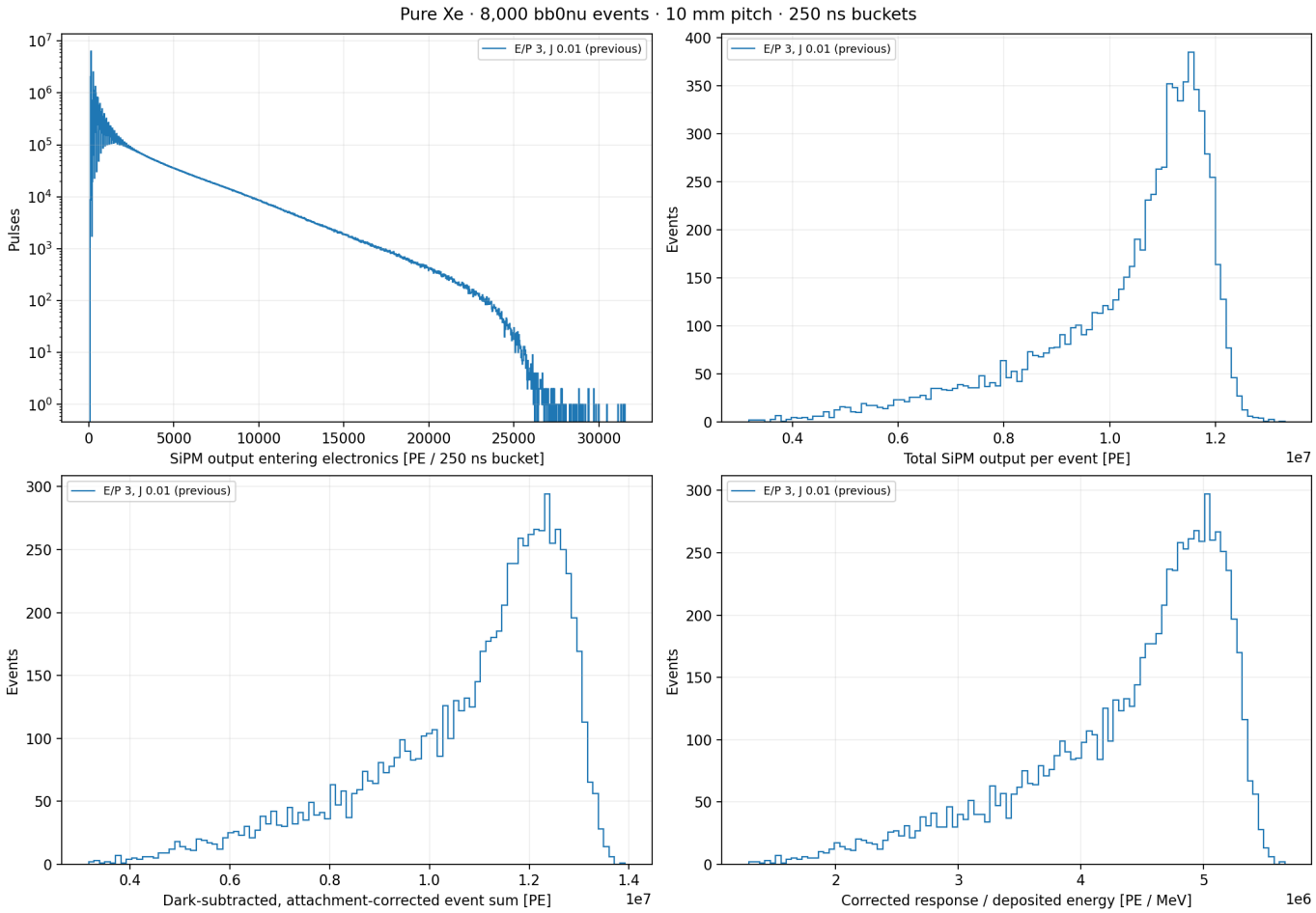
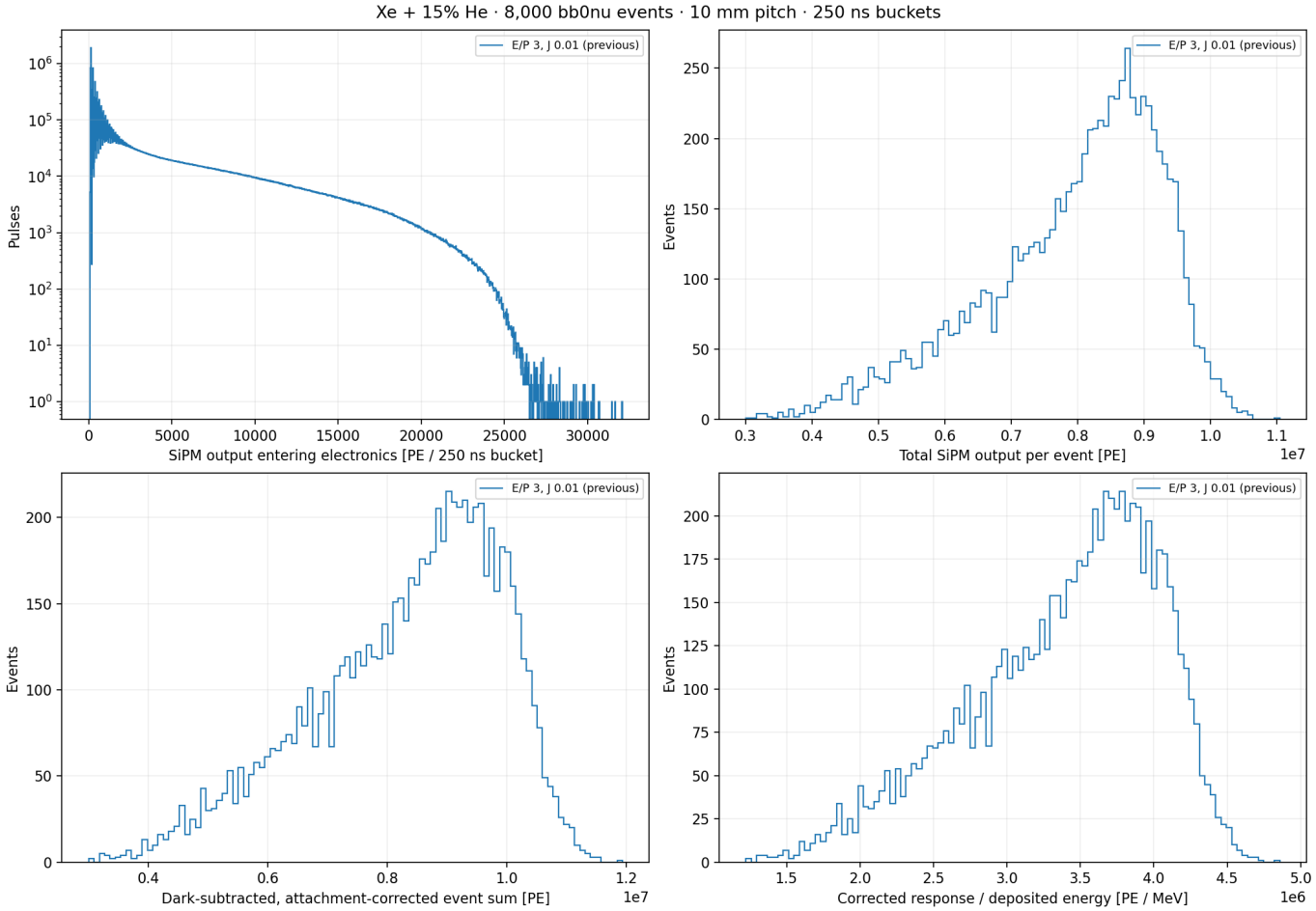
Electron operating points: signal entering electronics

NAUSICAA

Completed productions shown: legacy. Awaiting completion: ep2p5, ep3. Pressure 15 bar; holes 5 mm in diameter and 5 mm deep; PDE 0.3; gain 2.55e+06; 7854 microcells; recovery time 73.4 ns. The signal injected into electronics is the SiPM avalanche charge divided by the fully recovered single-PE charge. It includes microcell recovery and dark avalanches, before the 0.5 PE electronic noise and 1 mA clipping. The inclusive pulse distribution uses photon-positive SiPM–bucket cells. Event sums use every acquired cell. Expected stationary dark charge is subtracted before multiplying by exp(drift time / 15 ms). The dark baseline uses the stationary dark-only recovery factor; illumination can further reduce dark avalanche charge. The corrected sum retains SiPM nonlinearity. The tabulated width is the RMS divided by the mean of the complete event distribution. The response per deposited MeV additionally removes variations in the simulated deposited energy. Gaussian-equivalent FWHM is 2.355 times RMS; the distribution may be non-Gaussian. Both gases use the user-specified Oliveira Figure 6 readings: J = 0.015 at E/P = 2.5 and J = 0.025 at 3. The previous E/P = 3, J = 0.01 production is retained as a reference.

Gas	E/P	J	Events	Mean PE	RMS (%)	FWHM (%)	PE/MeV FWHM (%)	Clipped (%)	Retained (%)
Pure Xe	3.000	0.010	8000	1968.278	18.370	43.258	43.226	55.330	23.111
Xe + 15% He	3.000	0.010	8000	3552.795	18.896	44.497	44.467	65.568	13.854


Pure Xe · 8,000 bb0nu events · 10 mm pitch · 250 ns buckets
E/P 3, J 0.01 (previous)
Pulses
SiPM output entering electronics [PE / 250 ns bucket]
Events
Total SiPM output per event [PE]
1e7
Dark-subtracted, attachment-corrected event sum [PE]
1e7
Corrected response / deposited energy [PE / MeV]
1e6


Xe + 15% He · 8,000 bb0nu events · 10 mm pitch · 250 ns buckets
E/P 3, J 0.01 (previous)
Pulses
SiPM output entering electronics [PE / 250 ns bucket]
Events
Total SiPM output per event [PE]
1e7
Dark-subtracted, attachment-corrected event sum [PE]
Corrected response / deposited energy [PE / MeV]
1e6